ECEN 5692 Project | Matt Hartnett

$X(t) = \sum_l X_l u(t - l t_b)$

$u(t)$ must not extend for longer than $t_u = 40\ \mu s$

Pulse shape designed w/ BPSK

1) $X_K = 1$

1.1) $r_u(\tau) = \langle u(t+\tau), u(t) \rangle$

$\vec{y} = [\langle y(t), u_l(t) \rangle]_l$
$= R_u \vec{x} + \vec{n}$

$y_l = X_l r_u(l,l) + \sum_{l_1 \neq l} X_{l_1} r_u(l_1, l) + n_l$

Noise is zero mean, making the effective signal power:

$$\left| \underbrace{X_K r_u(K,K)}_{e_u} + \sum_{l_1 \neq K} X_{l_1} \mathrm{Re}\{r_u(l_1, K)\} \right|^2$$

$$\left| \frac{X_K}{e_u} \left(1 + \sum_{l_1 \neq K} \frac{X_{l_1} \mathrm{Re}\{r_u(l_1, K)\}}{X_l e_u} \right) \right|^2$$

w/ $X_K = 1$:

$$\left| \frac{X_K}{e_u} \left(1 + \sum_{l_1 \neq K} \frac{X_{l_1} \mathrm{Re}\{r_u(l_1, K)\}}{e_u} \right) \right|^2$$

$$\cancel{\boxed{\left| \frac{X_K}{e_u} \left(1 + \sum_{l_1 \neq K} \frac{X_{l_1} \mathrm{Re}\{r_u(l_1, K)\}}{e_u} \right) \right|^2}}$$

(1.2) $$\geq \left| e_u \left(1 - \sum_{l_1 \neq K} \frac{X_{l_1} \mathrm{Re}\{r_u(l_1, K)\}}{e_u} \right) \right|^2$$

worst case is $(1 - \Sigma)$ of all other symbols

Worst case:

$$\boxed{-20 \log_{10} \left(1 - \sum_{l_1 \neq K} \frac{X_l \mathrm{Re}\{r_u(l_1, K)\}}{e_u} \right) dB}$$

1.2) For worst case, $X_l \mathrm{Re}\{r_u(l_1, K)\}$ must be as negative as possible for $X_K = +1$

~~Therefor, X_l must always be~~

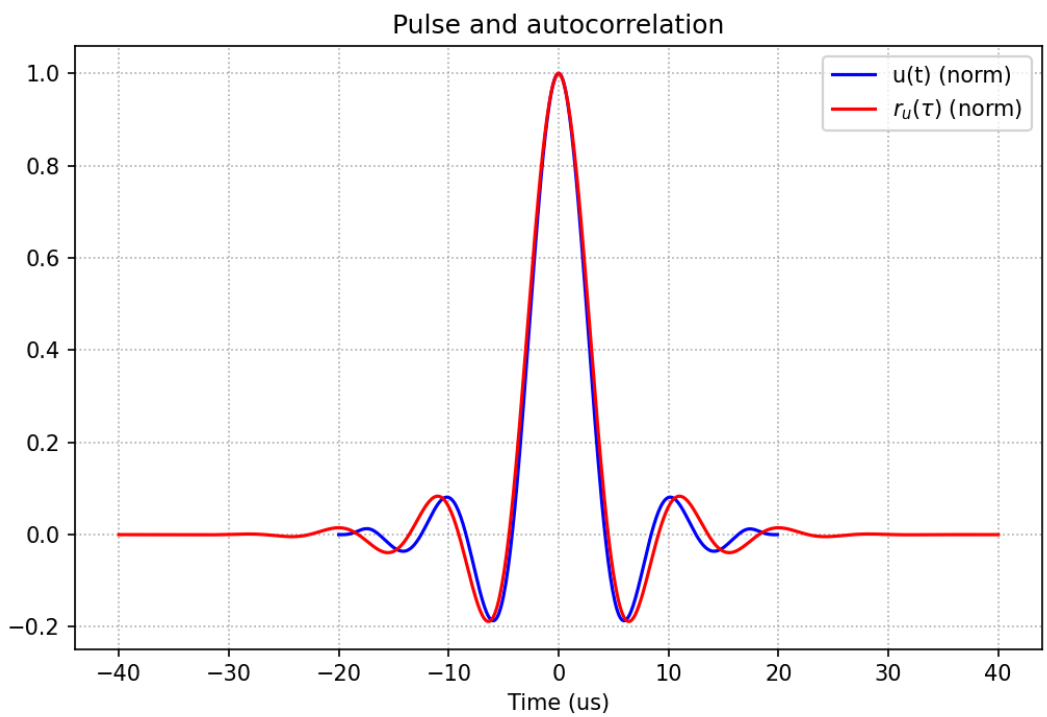
$$\boxed{X_l = \begin{cases} 1 & r_u(l_1, K) < 0 \\ -1 & r_u(l_1, K) \geq 0 \end{cases}}$$

2:
Pulse performance:
Max symbol rate = 219435 symbols/s
ISI degradation = 0.2480 dB

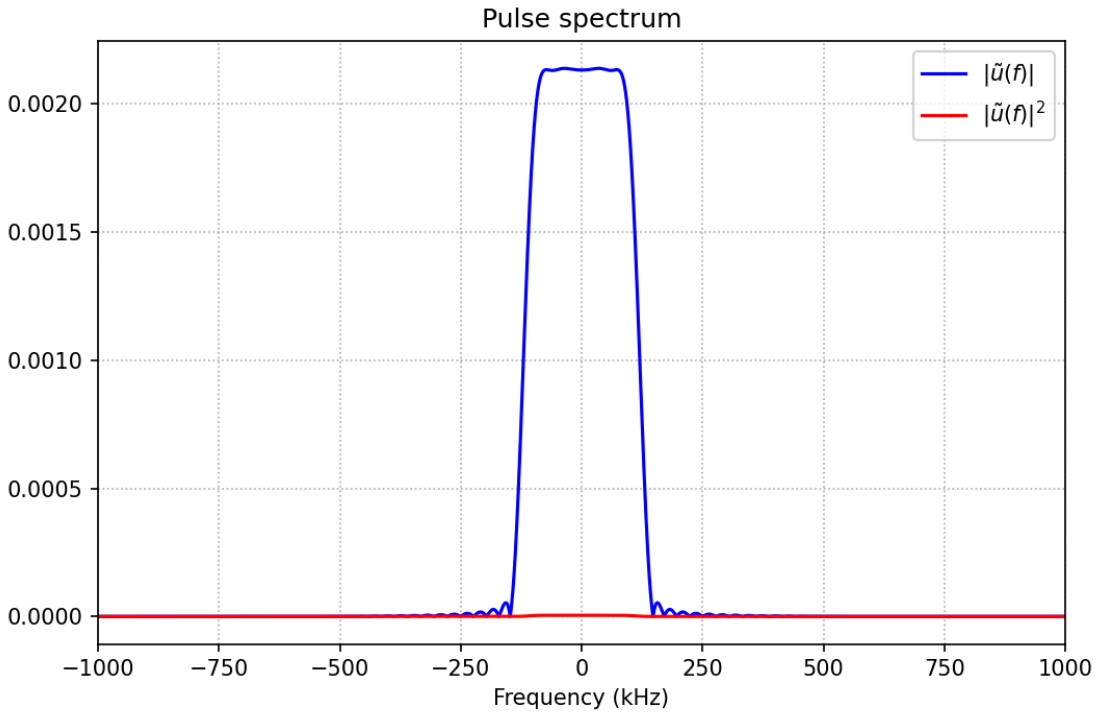
2.1:
Here are my design steps:

1. Infrastructure, constants, and other basic setup stuff that I know I'd need later in the project.
2. I created several helper functions to speed things up for me. I knew that I'd be building very similar pulses with slightly parameters, so I created functions that took an RCC pulse and modified with different parameters. I also added some checking functions to validate that my pulse met the specifications.
3. In optimization, I usually prefer a coarse then fine search, it usually helps speed things up a lot. The coarse search is more broad with larger steps in each parameter tuning loop, but it isn't the final result.
4. The fine search starts with the end of the coarse search. The coarse search returns a set of parameters, and the fine search checks +/- 1 step around each of those parameters to get a better value.
5. Once the best parameters have been found, the final step is to plot the pulse in the following ways.

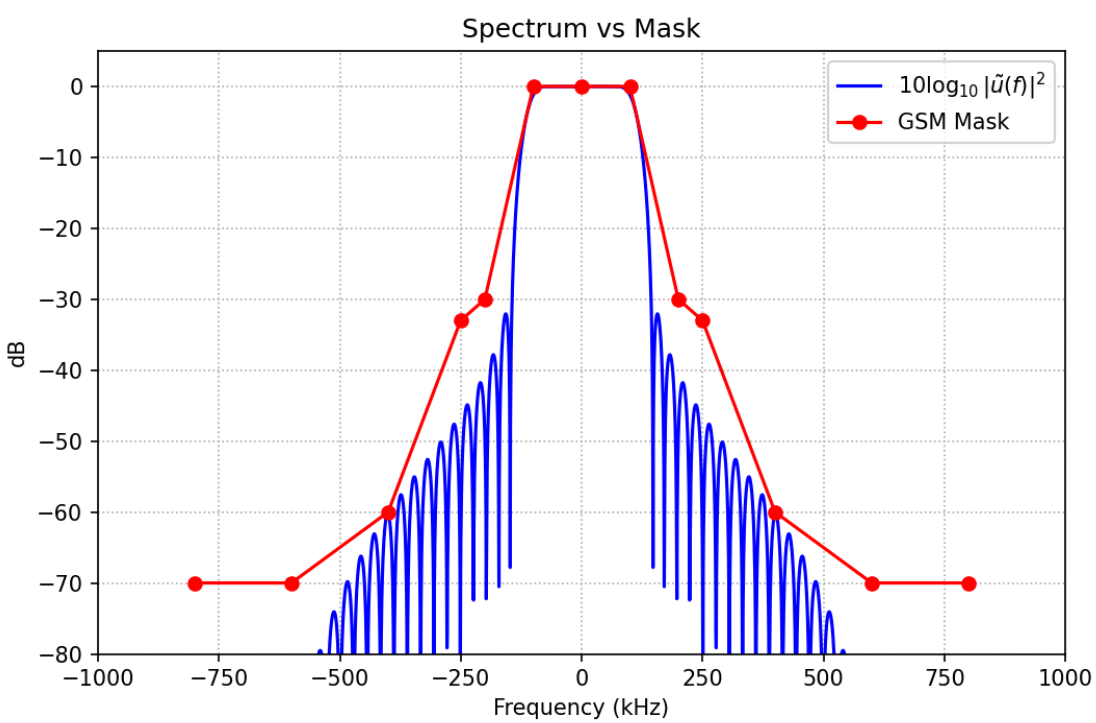
2.2:
u(t) and ru(τ):



u˜(f) and|u˜(f)|^2:

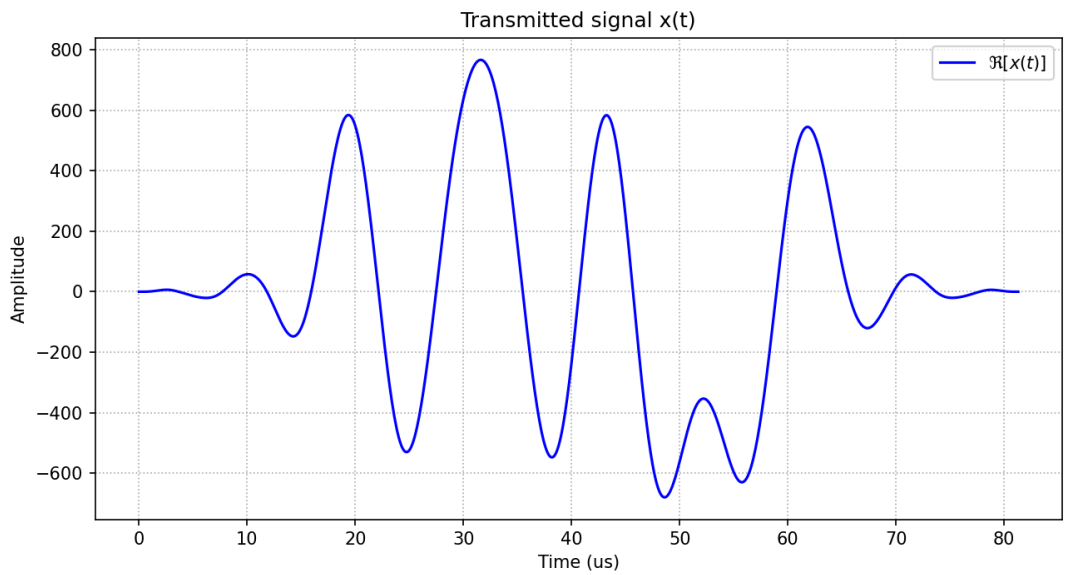
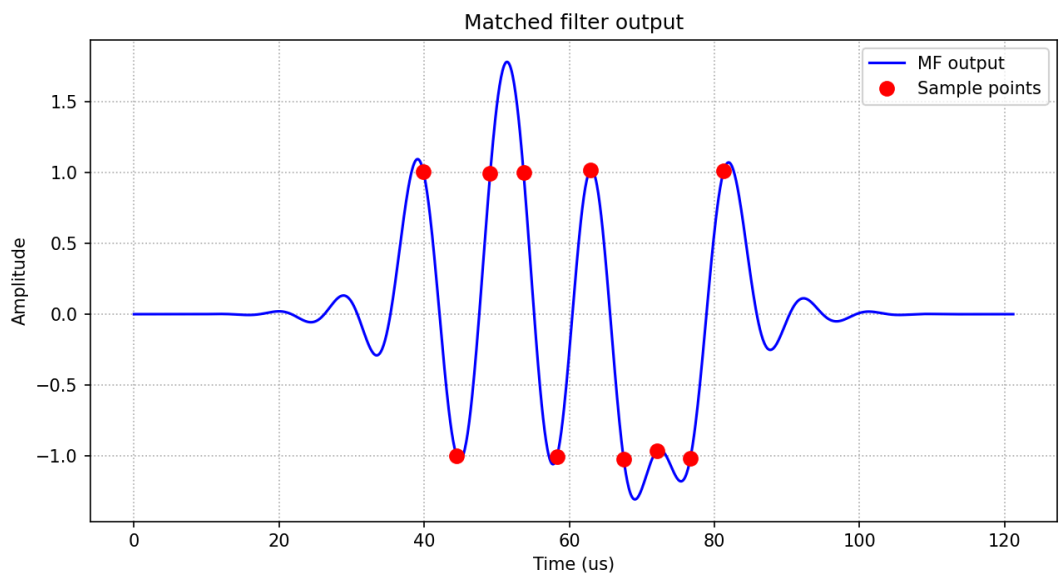


10 log10 (|u˜(f)|^2) and Mask(f):



2.3:

Transmitted envelope x(t):



Matched filter output: